

ELEC 7410 Solution: Homework Assignment #7 Nov. 13, 2025

1. **(Problem 5, Chapter 10, Gubner.)** Given $E\{X_t\} = 0$ and $E\{X_{t_1}X_{t_2}\} = R_X(t_1, t_2) = \min(t_1, t_2)$. Also, X_t is Gaussian, therefore, $X_t \sim \mathcal{N}(0, \sigma^2)$ with $\sigma^2 = E\{X_t^2\} = R_X(t, t) = t$. The pdf of X_t is

$$f_{X_t}(x) = \frac{1}{\sqrt{2\pi t}} e^{-\frac{x^2}{2t}}.$$

2. **(Problem 16, Chapter 10, Gubner.)** Given that the random sequence $\{X_n\}$ is WSS, implying that

$$E\{X_n\} = \mu \quad \text{and} \quad E\{X_n X_m\} = R_X(n - m) \quad \text{for any } n, m = 0, \pm 1, \pm 2, \dots$$

With $Y_n = X_n - X_{n-1}$, we have

$$\begin{aligned} E\{Y_n\} &= E\{X_n\} - E\{X_{n-1}\} = \mu - \mu = 0, \\ E\{Y_n Y_m\} &= E\{X_n X_m\} - E\{X_n X_{m-1}\} - E\{X_{n-1} X_m\} + E\{X_{n-1} X_{m-1}\} \\ &= 2R_X(n - m) - R_X(n - m + 1) - R_X(n - m - 1). \end{aligned}$$

Thus $\{Y_n\}$ is WSS since $E\{Y_n\}$ does not depend on n and $E\{Y_n Y_m\}$ depends on n and m only through the difference $n - m$.

3. **(Problem 52, Chapter 10, Gubner.)** Let $R(\tau) \leftrightarrow S(f)$, i.e., $S(f) = \int_{-\infty}^{\infty} R(\tau) e^{-j2\pi f\tau} d\tau$, and let $\bar{R}(\tau) \leftrightarrow \bar{S}(f)$ where $\bar{R}(\tau) = R(\tau - \tau_0) + R(\tau + \tau_0)$. We have ($j = \sqrt{-1}$)

$$R(\tau - \tau_0) \leftrightarrow S(f) e^{-j2\pi f\tau_0} \quad \text{and} \quad R(\tau + \tau_0) \leftrightarrow S(f) e^{j2\pi f\tau_0}.$$

Therefore,

$$\bar{S}(f) = S(f)(e^{-j2\pi f\tau_0} + e^{j2\pi f\tau_0}) = 2S(f) \cos(2\pi f\tau_0).$$

Since $R(\tau)$ is a correlation function, $S(f)$ is a power spectral density (PSD) function which is equivalent to $S(f) \geq 0$ for $-\infty < f < \infty$ and $S(f) = S(-f)$. For $\bar{R}(\tau)$ to be a correlation function, $\bar{S}(f)$ should be a PSD function. Notice that $\cos(2\pi f\tau_0)$ is a periodic function in f with (fundamental) period $F = \tau_0^{-1}$, and it is ≥ 0 for $-\frac{1}{4\tau_0} \leq f \leq \frac{1}{4\tau_0}$ but < 0 for $\frac{1}{4\tau_0} \leq f \leq \frac{3}{4\tau_0}$, and so on.

So for an arbitrary $S(f)$, although $\bar{S}(f) = \bar{S}(-f)$, for some values of f (such as $\frac{1}{4\tau_0} \leq f \leq \frac{3}{4\tau_0}$), $\bar{S}(f) < 0$. This rules out (a). But if we pick $S(f) = 0$ **outside** the interval $-\frac{1}{4\tau_0} \leq f \leq \frac{1}{4\tau_0}$, then $\bar{S}(f)$ is a valid PSD function, therefore, equivalently, $\bar{R}(\tau)$ is a correlation function. This satisfies the assertion in (b), which, in turn, rules out (c).

4. **(Problem 25, Chapter 11, Gubner.)** Given: X_t is WSS, $E\{X_t\} = 0$, $E\{X_t X_s\} = \sigma^2 \delta(t - s)$. Define

$$V_t = \int_0^t X_\tau d\tau$$

For $0 \leq s < t < \infty$:

$$E\{V_t V_s\} = \int_0^t \int_0^s \underbrace{E\{X_\alpha X_\beta\}}_{\sigma^2 \delta(\alpha-\beta)} d\alpha d\beta = \int_0^{\min(t,s)} \sigma^2 d\beta = \sigma^2 s \text{ since } s < t,$$

where we have used the fact

$$\int_0^s \sigma^2 \delta(\alpha - \beta) d\alpha = \begin{cases} \sigma^2 & \text{if } 0 < \beta < s \\ 0 & \text{otherwise} \end{cases}$$

5. **(Problem 27, Chapter 11, Gubner.)** W_t is a Wiener process with $E\{W_t^2\} = \sigma^2 t$, $t \geq 0$. Let $Y_t = e^{W_t}$. For $t_2 > t_1$,

$$\begin{aligned} R_Y(t_1, t_2) &= E\{Y_{t_1} Y_{t_2}\} = E\{e^{W_{t_1} + W_{t_2}}\} = E\{e^{W_{t_1} + W_{t_2} - W_{t_1} + W_{t_1}}\} \\ &\stackrel{\text{independent increment}}{=} E\{e^{2W_{t_1}}\} E\{e^{W_{t_2} - W_{t_1}}\}. \end{aligned}$$

If $X \sim \mathcal{N}(0, \sigma^2)$, then

$$E\{e^X\} = \int_{-\infty}^{\infty} e^x \frac{1}{\sqrt{2\pi}\sigma} e^{-x^2/(2\sigma^2)} dx.$$

The exponents in the integral above can be “simplified (complete the square)” as

$$\frac{1}{2\sigma^2} (x^2 - 2\sigma^2 x + \sigma^4 - \sigma^4) = \frac{1}{2\sigma^2} (x - \sigma^2)^2 - \frac{\sigma^2}{2}.$$

Thus

$$E\{e^X\} = e^{\sigma^2/2} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\sigma^2)^2/(2\sigma^2)} dx = e^{\sigma^2/2}.$$

Now $2W_{t_1} \sim \mathcal{N}(0, 4\sigma^2 t_1)$ and $W_{t_2} - W_{t_1} \sim \mathcal{N}(0, \sigma^2(t_2 - t_1))$. Therefore,

$$R_Y(t_1, t_2) = e^{2\sigma^2 t_1} e^{\sigma^2(t_2 - t_1)/2} = e^{\sigma^2(t_2 + 3t_1)/2}.$$

6. **(Problem 28, Chapter 11, Gubner.)** Given Wiener W_t with $E\{W_t^2\} = \sigma^2 t$, $t \geq 0$. Let

$$\mathbf{X} = \begin{bmatrix} W_{t_1} \\ W_{t_2} \\ \vdots \\ W_{t_n} \end{bmatrix}$$

where $0 < t_1 < t_2 < \dots < t_n$. Since $E\{\mathbf{X}\} = \mathbf{0}$, $\text{cov}(\mathbf{X}) = E\{\mathbf{X}\mathbf{X}^\top\}$. Since $E\{W_{t_\ell} W_{t_m}\} = \sigma^2 \min(t_\ell, t_m)$, we have

$$\text{cov}(\mathbf{X}) = \sigma^2 \begin{bmatrix} t_1 & t_1 & t_1 & \cdots & t_1 \\ t_1 & t_2 & t_2 & \cdots & t_2 \\ t_1 & t_2 & t_3 & \cdots & t_3 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ t_1 & t_2 & t_3 & \cdots & t_n \end{bmatrix}$$